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Robust classification model for identifying stroke patients utilising a machine learning-based ensemble stacking method

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Abstract

Purpose of the study: Cerebrovascular accident (CVA) occurs quickly and disrupts brain blood flow, causing neurological problems. Risk factors include smoking, diabetes, hyperlipidemia, hypertension, and atrial fibrillation. Many brain stroke prediction systems are computationally demanding, sluggish, and unreliable. This study uses interpolation spline curves that impute data and Principal Component Analysis to identify the most important factors for stroke prediction. Using stack ensemble machine learning methods, the goal is to improve the precision of stroke disease identification. **Used method:** Using interpolation spline curve data imputation and Principal Component Analysis to preprocess and extract EHR data is unique. In addition, this article also proposes a novel stacking ensemble technique using bagging and boosting as foundation learners. Bagging and boosting enhance prediction by reducing variation and bias. Stack-based ensemble learning classifies brain strokes. Build base and meta-learners with Random Forest, XGBoost, Decision tree classifier, KNN, and Logistic Regression. The evaluation of the efficiency of the machine learning model determines the optimal stacking ensemble meta-learner model. **Brief Description of Results:** The results emphasize the best meta-learner model of the stacking ensemble and the impact of the imputation technique on the representation of the electronic health record. Using PCA, the most important factors have been identified for detecting stroke. The results of the analysis showed that the logistic regression as a meta-educator had the best predictive performance, achieving 0.96 precision and 0.98 recall value, respectively, with 97% accuracy and AUC curve value of 0.99. **Findings:** The Decision Tree Classifier and K-Nearest Neighbor as a metalearner got 95%, 96% accuracy and both 0.95 AOC curve values in this data set, showing that the method works. Precision values of 0.93, and 0.95 for the Decision Tree Classifier and KNN Classifiers as meta-learner confirm their robustness. These findings suggest that interpolation spline curve and stacking-based ensemble machine learning can improve the identification of brain stroke from incomplete electronic health records.

1. Introduction

Advancements in technology are leading to an increase in the average lifespan of humans. One of the predictors of physical inactivity is the increasing use of technology in daily life, such as computers, cellphones, and portable gadgets. In addition to the aging population, the younger generation is also facing various health problems as a result of their poor lifestyle, such as diabetes, hypertension, cardiovascular disease, and strokes. To track people's health using biomarkers and non-intrusive smart techniques, smart medical equipment is necessary [1].

In recent years, stroke has been a significant problem that affects people's health. A stroke, often referred to as a cerebrovascular accident, is a neurological condition that can be caused by ischemia or hemorrhage of the brain arteries. Typically, it results in a variety of physical and cognitive deficits that impede function [2]. Every

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year, almost 16 million people have a stroke, which is associated with high socioeconomic costs. The use of machine learning has advanced and expanded tremendously in recent years in a variety of healthcare systems. Blood clotting is the primary manifestation of stroke, which disrupts blood flow to brain neurons. There are several stroke levels. Although major strokes can be deadly, minor strokes affect the blood supply to specific parts of the brain. A stroke is a medical emergency that needs to be handled with care. Moving difficulties, disorientation, poor verbal communication, and difficulty understanding are common symptoms. Stroke causes permanent neurological damage and death [3]. Ischemic and hemorrhagic strokes are the two types of strokes. An ischemic embolic stroke occurs when a blood clot narrows the arteries that supply the brain rather than the brain itself. Approximately 13% of stroke cases are hemorrhagic strokes. The rupture of a cerebral blood artery causes these strokes. The most common type of stroke is ischemic. Hemorrhagic strokes are also called intracerebral hemorrhages, or ICH. An ICH occurs when a blood artery breaks and blood accumulates in the tissue surrounding the artery. This causes blood loss in surrounding tissues and puts pressure on the brain. Getting medical assistance as soon as possible is essential for the best chance of recovery. Prevention is also very important. If you control your risk factors, you can greatly reduce your chance of having a stroke of any type [4].

In India, stroke is becoming more prevalent; it is currently the fifth most common cause of disability and the fourth leading cause of death. Previous research has suggested that the annual incidence of stroke in India is between 105 and 152 incidents per 100,000 people [5]. Stroke affects an estimated 800,000 Americans each year. Since 600,000 people have had strokes for the first time, 200,000 more will get one. This suggests that in the US, a stroke occurs every 40 seconds. The stroke death rate is 3.5 minutes (CDC web page) [6]. Stroke deaths have been 10 times higher in developing nations since 1970, and are expected to treble globally by 2030. Magnetic resonance imaging (MRI) and computed tomography (CT) are commonly used in studies to identify strokes. Furthermore, due to the excessive costs associated with MRI and CT scans, people in underdeveloped nations cannot access the benefits of these diagnostic instruments [7]. Although MRI does not involve ionizing radiation and is generally safer in terms of allergic reactions, its high cost and limited availability can hinder its use in certain regions. CT scans, though more accessible, involve ionizing radiation and can pose risks to some patients, such as allergic reactions to contrast agents. In this paper, we present a system for the identification of brain stroke patients which is based on filling missing data algorithms and machine learning models. To achieve this goal, we experiment with several machine learning models and fill in missing values strategies to attain this goal. The key contributions of this work are summarized below:

- Preprocessing of data including imputation of missing values using mean and interpolation spline curves. Also, solve the problem of data imbalance.
- Analyze the loadings (eigenvectors) of each principal component in order to determine the primary contributions of the original characteristics to each new component.
- We propose a new framework for brain stroke identification that takes advantage of both bagging and boosting as base learners, combining their complementary strengths to enhance model performance and compare with a single classifier.
- Compare the performance of various classifiers after employing different data imputation techniques such as mean and interpolation spline curves.

The remainder of the paper is organized as follows. Section 2 presents a comprehensive review of the literature. Section 3 will describe the methodology used. In sections 4 and 5 a discussion of the results and conclusion will be presented.

2. Literature survey

This section presents the literature survey of various studies on stroke classification.

Shanthi *et al* forecast thromboembolic stroke disease using ANN [8]. In this investigation patients with stroke symptoms are used. Input feature selection is performed stepwise backward. Removal of irrelevant input improves the performance of the neural network. This study shows better accuracy in predicting stroke diseases based on ANN to 89% with greater consistency. ANN predicts stroke well.

The primary result of several health-related data mining articles was the development of decision support systems. Sheng-Feng Sunga *et al* [9] conducted an analysis of data from patients with acute ischaemic stroke to create a predictive model to determine the severity of the condition. The authors used the K-nearest neighbor model, multiple linear regression, and regression tree model in their investigation. These models achieved accuracies of 0.743, 0.742, and 0.737, respectively, with a confidence interval 95%.

Retrospective data from adult patients registered in THIN between 2007 and 2011 and who had a stroke were used in [10] to identify potentially undiagnosed post-stroke cases in the United Kingdom. This study examined healthcare resources, socioeconomic status, BMI, age, and stroke history in 45,613 clinical health record samples. Data included stroke records and 12-month pre and poststroke data. To forecast stroke-related spasticity. The best data- and clinician-trained prediction method was used to detect PSS patients (stroke event with a spasticity-related Read code within 12 months after stroke). Random forest, CART, and ADABOOST classified. Random forests trained in data work better.

The authors [11] developed a model to predict post-stroke nemonina using a deep learning model. In this work, data has been collected from acute ischaemic stroke in hospitals. Multiple layer perceptron (MLP) neural networks and recurrent neural network (RNNs) (i.e., attention-augmented gated recurrent unit (GRU)) has implemented and compare with other classical machine learning algorithms. Results shows that the attention-augmented GRU model achieved gives best performance compare to other machine learning model.

In [12] studied stroke detection accuracy. Their main goal was to use machine learning to predict brain strokes. This study uses Kaggle physiology data from 5,110 people. The 12 characteristics include biological and demographic data. The preprocessing method fills the null values of the BMI column with the mean of the data column. Data preprocessing balances the data for correctness. To balance the data, SMOTE was used. Logistic regression (LR), Decision Tree classification (DT), Random Forest classification (RF), and Voting classification were used to predict accuracy, with Random Forest performing best.

In [13] used electronic health records to determine the type of stroke that can occur or occur from a person's physical state and the data from the medical report. They have collected data from the hospital. Their database contains string values; therefore, they used integer encoding during the preprocessing stage. There are 28 attributes used for stroke detection. In their study, some of the attributes are missing in the dataset, so they replaced the missing value with 0. Machine learning algorithms, such as Naive Bayes, J48, k-NN, and Random Forest, are used to classify the different types of strokes. In their work, J48, k-NN, and Random Forest performed well in identifying different types of strokes.

In [14] designed a model for the early prediction of stroke. In their study, the appearance of hypertension, body mass index level, heart disease, mean glucose level, smoking status, previous stroke, and age of the person were used as attributes. In their work, multiple classifiers are aggregated using the weighted voting approach to achieve the highest accuracy.

In [15] designed a model for the analysis of stroke prediction. In their experiment, 5,110 observations with 12 attributes of physiological health records were used. During pre-processing, authors drop the id column. The authors used the synthetic minority oversampling technique (SMOTE) after preprocessing the datasets in the EDA step to improve the model accuracy. For classification, Naive Bayes, Random Forest, Ada Boost algorithm used for the stroke prediction analysis, and Random forest give better results than other machine learning algorithms in their work.

Authors, in [16] performed research to analyze the performance of stroke prediction. The data set used was taken from the kaggle and 12 biological and demographic attributes were used. In data preprocessing steps, the author replaces the null value with the mean value of the column. To improve the accuracy of the model they handled the imbalanced data by using a sub-sampling technique. For classification, the authors used logistic regression, decision tree classification, random forest classification, closest neighbors K, support vector machine, and naive Bayes. In their work, KNN and SVM give better results.

In [17] have implemented a machine learning model to predict the severity of brain stroke. A total of 4,799 subjects were included in the study, including 3,123 men and 1,676 women, along with an overview of the key characteristics. In the data pre-processing part, the author removed the 'Gender' attributes. Decision Tree, Gaussian Naïve Bayes, Logistic Regression, Linear SVM, AdaBoost, etc for the classification task. In their work, Random Forest achieves the highest accuracy compared to other machine learning algorithms for the prediction of brain strokes.

The research carried out by [18] presents the data mining classification technique for the prediction of stroke. The researchers used data obtained from the King Abdulaziz Medical City (KAMC) data governance department for their investigation. Their database contains 1,004 attributes. To reduce the dimensionality of the feature vector, they used PCA. They applied MLP, JRIP, and C4.5. for the classification task. In their research, Jrip gives better performance without PCA and MLP gives better performance after applying PCA.

In [19] proposed an early stage hybrid supergroup learning model for the prediction of diabetes risk. The superlearner model was constructed using four base-learner support vector machines, a metalearner as a consequence of an investigation. The robustness of the proposed model was assessed using three different datasets. The chi-square method was identified as the most effective way to select characteristics among five different methods. Additionally, hyper-parameter values were adjusted using GridSearch. The innovative proposed super-learner model achieved superior accuracy in the identification of diabetic complications compared to base-learners for the prediction of early-stage risk of diabetes.

In [20] proposed a stack-based ensemble learning model.This work uses FIFA'19 game data to categorize footballer positions using a stack-based ensemble ML model. This is achieved by a two-stage application. Four feature selection algorithms pick 10 features in the initial step. The second stage uses Deep Neural Networks, Random Forest, Gradient Boosting, and Logistic Regression as single-based algorithms and meta-learners in the stacked model. The Chi-square feature selection approach and the stacked-based ensemble learning model produced the best accuracy.

Table 1 highlights current related approaches and efforts to predict brain stroke. The strengths of the studies provided include the use of machine learning algorithm models such as Random Forest, AdaBoost, and ensemble models for accurate prediction and treatment of imbalanced data using SMOTE techniques. In order to eliminate multicolinear features, some studies used correlation analysis. As a feature selection technique, some research used the chi-square test, information gain, and mutual information. Some studies also used principal component analysis (PCA) to perform dimensionality reduction, making models faster and with maximum efficiency. Nevertheless, it has limitations such as specificity of the datasets used, low external validity, and generalizability of the phenomenon, which is highly questionable in most of the real world situations. Most of them also give methods and results for a particular type of prediction task, i.e., severity or spaticity only, and not for other related tasks. To overcome these restrictions, some researchers started using ensemble learning, where multiple algorithms were implemented simultaneously, used a high level of preprocessing, including interpolation splines, or trained the model with the clinician's input. The proposed changes increase the stability of the models and their relevance to various contexts within the healthcare domain.

The gaps identified in the papers are also the following: Firstly, many of the studies used specific datasets without testing on other types of dataset or real ones. Most studies did not address realism by failing to consider issues related to noisy or incomplete data typical in the healthcare scenario. Some articles focused on a single aspect of the rehabilitation process, such as the severity or spasticity of the presenting stroke, and did not include a detailed discussion of general prediction methods. There was fewer usage of complex deep learning variations compared to more primitive models of artificial intelligence. In addition, there has been little cross-comparison of different algorithms to determine which methodology was the best to use, and most of the articles had poor assessments of models using measurement standards other than accuracy, such as recall value or F1 scorewise. Finally, none of the above works provided ideas for the implementation of these models into operational applications for use in healthcare organizations in real-time settings.

3. Proposed work

The following proposed methodology is presented. Figure 1 gives a graphical presentation of the proposed methodology.

3.1. Dataset

The 'Health-stroke-data' is a Kaggle dataset [23] for predicting strokes. There are 5,000 observations in the dataset, each with a different attribute. The main features of the data set include the categorical variables in the overview, including marital status, work type, id, sex, hypertension, heart disease, residence type, smoking status, and the patient's prior history of stroke. The average glucose (sugar) level, age, and BMI are qualitative variables.

3.2. Pre-processing

Data preprocessing is necessary before creating a model to eliminate unwanted noise and outliers from the dataset that might cause the model to deviate from its intended training. The relevant dataset must first be gathered, and then the data needs to be cleaned and ready for model development. The dataset used has twelve characteristics, as mentioned previously. First, the column id is left out because it does not affect how the model is built. The dataset is then checked for null values and any found are filled in. In this instance, the interpolation cubic spline is used to fill in the null values in the BMI column.

Label encoding transforms the dataset's string literals into understandable integer values. Strings must be converted to integers because the computer is frequently trained in numbers. The dataset comprises five attributes that are classified as string data types. In label encoding, each piece of text in a dataset is turned into an integer, which turns the dataset as a whole into a set of numbers. The sample used to predict strokes is very unequal. There are a total of 5,110 records in the collection. Only 249 of the records show that there is a chance of having a stroke, while 4,861 show that there is no chance of having a stroke.

The missing entries in the dataset must be filled in as the second component of the issue. Several approaches were used, including hot-deck imputation, column mean, and column median filling for missing features. They

Table 1. Summary of the related work.

References	Objective	Approach	Strength	Limitation
[8].	To predict thrombo-embolic stroke using Artificial Neural Networks	Artificial Neural Network (ANN) with stepwise backward input feature selection to remove irrelevant features	Effective feature selection improved ANN performance	Limited to specific input features and dataset; generalizability not tested.
[9].	To develop a severity index for acute ischemic stroke using data mining techniques	Data Mining with K-Nearest Neighbor (KNN), Multiple Linear Regression and Regression Tree Model.	Comprehensive analysis of multiple models for severity prediction.	Retrospective data; limited by dataset representativeness.
[10].	To identify potentially undiagnosed post-stroke spasticity patients using predictive analysis.	Random Forest, CART preprocessed and AdaBoost prediction methods	Inclusion of clinician-trained methods improved robustness.	Focus on post-stroke spasticity only; reliance on. healthcare records.
[11].	To predict post-stroke pneumonia using electronic health records and deep learning.	Deep Neural Networks with preprocessed electronic health record data.	To predict specific complications.	Limited to pneumonia prediction post-stroke.
[12].	To detect and predict stroke disease using robust learning approaches	Random Forest, Decision Tree, Logistic Regression, and Voting Classification	Handled data imbalance effectively using SMOTE	No detailed accuracy comparison; data preprocessing specifics not elaborate.
[13].	To classify stroke types using EHR and machine learning algorithms.	Machine Learning Algorithms like Naive Bayes, J48, K-NN, and Random Forest.	Use of multiple classifiers provided robust analysis.	Data missingness handled with zero replacement; dataset features limited
[14].	To predict the occurrence of stroke using machine learning model.	Weighted voting using multiple machine learning algorithms.	Diverse classifier.	It does not examine the impact of each attribute on model performance
[15].	To analyze stroke prediction using multiple machine learning classifiers.	Naive Bayes, Random Forest, and AdaBoost with SMOTE for handling imbalance.	Use of multiple classifiers provided a comparative analysis.	Dataset features not elaborated upon; lack of external validation.
[16].	To evaluate the performance of stroke prediction using ML classification algorithms.	Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Naive Bayes	Inclusion of multiple classifiers showed varied strengths.	Sub-sampling technique used; impact on smaller datasets unclear.

Table 1. (Continued.)

References	Objective	Approach	Strength	Limitation
[17]	To predict the severity of brain stroke using machine learning models.	Decision Tree, Gaussian Naive Bayes, Logistic Regression, etc. Linear SVM, AdaBoost,etc	Detailed model comparison highlighted Random Forest's efficiency.	Limited generalizability to diverse datasets.
[18]	To predict stroke using data mining classification techniques.	MLP, JRIP, and C4.5 with PCA for dimensionality reduction.	Use of PCA improved MLP performance significantly.	Dataset constraints and limited evaluation.
[19].	To develop an early-stage hybrid supergrouped learning model for diabetes risk prediction	Four base learners (e.g., SVMs) with a metalearner using GridSearch for hyperparameter tuning.	Robust feature. selection using chi-square improved accuracy.	Application limited to diabetes risk prediction; not stroke-specific
[21].	To develop predictive models that are capable of precisely estimating the risk for mortality in individuals suffering from liver cirrhosis.	Principal Component Analysis for eatures extraction and Artificial Neural Network (ANN) for classification.	Concurrently achieved a high recall and high precision.	A common issue in imbalance datasets.
[22].	To compare and identify machine learning algorithms for image classifications problem with small dataset.	Multiple learning algorithms such algorithms such optimizer, AdaGrad optimizer, RMSProp optimizer, Adam optimizer.	The AdaGrad learning algorithm attains superior accuracy, reduced training duration, and diminished memory usage relative to other learning algorithms.	Focuses on small datasets.

are the benchmark techniques against which we would like to evaluate the other algorithms that we develop and are widely used in statistics.

3.2.1. Mean imputation

This algorithm is frequently used and is the most common method of replacing missing values with the mean of each column. This strategy is straightforward and easy to adopt, but has downsides. If the number of missing values is considerable, all of them are replaced by the same imputation value, that is, the mean, causing the distribution's shape to shift.

3.2.2. Data imputation by using interpolation cubic spline curve

The objective of cubic spline interpolation is to draw a smooth curve through a given set of points such that the slope and curvature are also continuous along the curve (such curve is known as spline). Let y = f(x) be the required spline where the function is defined in the domain [a,b]. If we divide the intervals into n subintervals such as $[x_0, x_1]$, $[x_1, x_2]$, …, $[x_{n-1}, x_n]$, where $a = x_0 < x_1 < x_2 < \ldots < x_{n-1} < x_n = b$, then the points $x_1, x_2, \ldots, x_{n-1}$ are called internal nodes.

The motive is to determine such a spline by interpolation that maintains continuity in the first and second derivatives, both inside the subintervals and at the interpolating nodes. This acquires a smoother interpolating function as a result. Because of the continuity of the first derivative, there will be no sharp corners on the graph y = f(x). The radius of curvature is defined at every point because of the continuity of the second derivative [24].

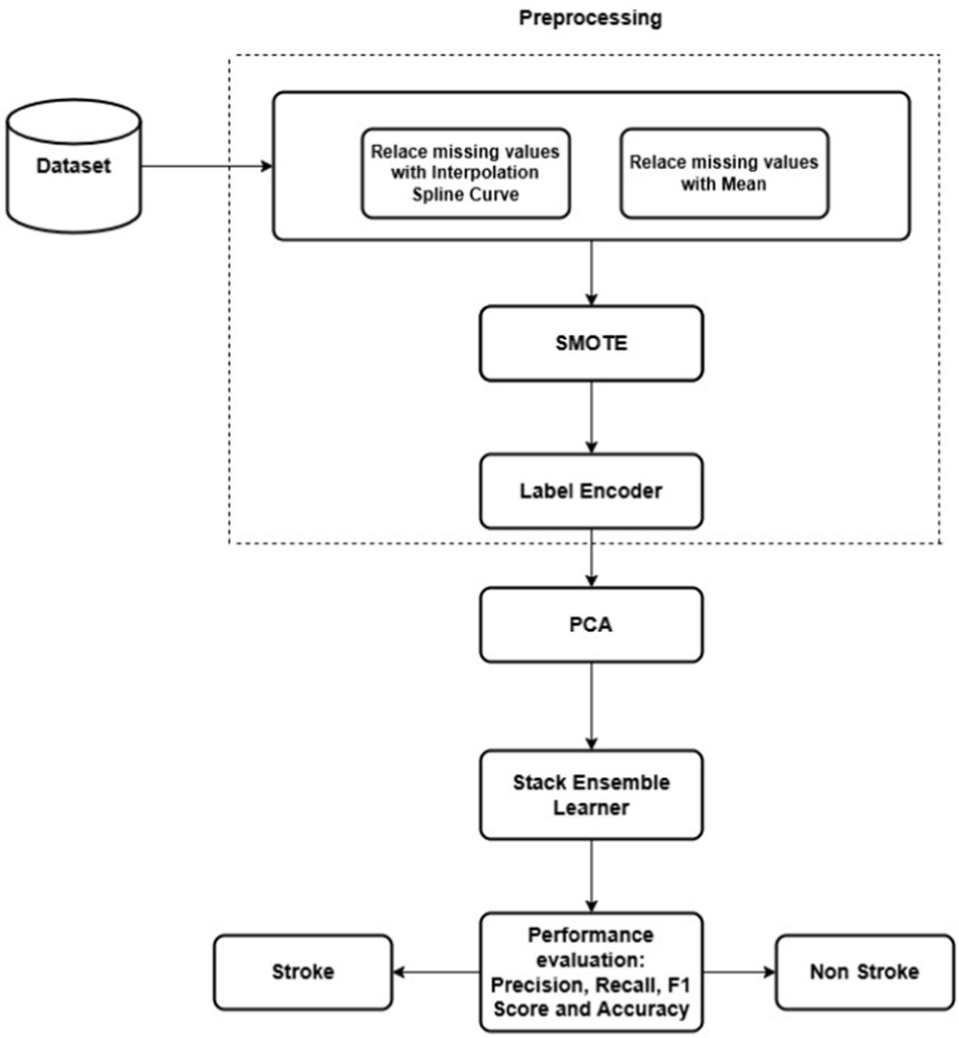


Figure 1. The step-by-step process of the proposed work.

To construct the spline, we start with the piecewise cubic spline which is defined in the i-th subinterval $[x_{i-1}, x_i]$ as,

$$f(x) = a_i + b_i x + c_i x^2 + d_i x^3;\ i = 1,\ 2,\ ..n \tag{1}$$

where a_i, b_i, c_i, d_i are the $4n$ coefficients for $i = 1, 2, \ldots, n$.

Being cubic polynomials, each of the functions f_i is a C^2 function in the i-th subinterval. The cubic spline interpolation in (1) must satisfy the following condition:

$$f_i(x_i) = f_{i+1}(x_i) \tag{2}$$

$$f_i'(x_i) = f_{i+1}'(x_i) \tag{3}$$

$$f_i''(x_i) = f_{i+1}''(x_i) \tag{4}$$

For cubic spline interpolation, we need to determine these 4n parameters. In this process, the following natural boundary conditions will be used:

$$f_1''(x_0) = f_n''(x_n) \tag{5}$$

3.2.3. SMOTE (Synthetic Minority Over-sampling Technique)

SMOTE (Synthetic Minority Oversampling Technique) is a powerful technique used to address the challenge of imbalanced datasets in machine learning. Class imbalance occurs when one class is significantly underrepresented compared to the others, leading to biased model performance. SMOTE specifically focuses on the minority class by generating synthetic samples to re balance the dataset [25]. The dataset used in this work

contains 4861 non-stroke records and 249 stroke records. To balance this imbalanced data set, the synthetic minority oversampling technique is used.

3.3. Principal component analysis

Reducing the dimensionality of the dataset from many parameters to a small set of values is the main use of Principal Component Analysis (PCA). Using a statistical method known as principal component analysis (PCA), a collection of potentially correlated variables can be transformed into a set of linearly uncorrelated variables. Principal component analysis (PCA) transforms the data into a collection of non-connected feature vectors. In this study, principal component analysis (PCA) was used to identify the best features to discriminate between stroke and non-stroke patients. Covariance matrices, eigenvectors, and eigenvalues are calculated [26].

The covariance was determined using the following formula:

$$C = Cov(X, Y) = \sum \frac{(X_i - X_m)(Y_i - Y_m)}{N} \tag{6}$$

where N is the number of subjects or samples in the data sets.
X_m indicates mean value of the sample in the initial set of variables.
X_i is the subject of the column ith in the first set of variables.
Y_i is the sample mean in the second set of variables.
Y_m is the subject column ith in the second set of variable
To diagonalize the dispersion matrix, compute the matrix of V eigenvectors as follows:

$$V^{-1}CV = D \tag{7}$$

Matrix D is created using the eigenvalues of matrix C. The columns of V form an orthogonal matrix with unit length, defining the primary components as linear combinations of data along the directions with zero covariance. The variance for every primary element is determined by the values on the diagonal of matrix D.

The contributions of each variable to the first two major components are presented in table 2. For each principal component, the square sum of all loadings is equal to unity. Consequently, a variable is said to have a significant contribution to the principal component if its loading reaches a threshold value of $\sqrt{\frac{1}{10}} = 0.31$. It is evident from this that the first main component is strongly influenced by the following variables: age, average glucose level, BMI, and type of work.

The first two principal components explain 30% and 15% of the variance in the stroke dataset, respectively, capturing 4% of the total variance. This indicates that a substantial amount of information can be retained in a 2D space. The components help identify key characteristics shown in figure 2 that influence stroke prediction and improved model performance.

3.4. Classification

3.4.1. Bagging

Bagging is a technique used to reduce the variation in a decision tree while preserving its bias and minimizing overfitting. The Bagging Tree creates numerous sets of input data from training samples by randomly selecting and replacing them. The selected subset of data is used to train the designated trees and produce models. Afterwards, the average of all forecasts from these trees is used to determine the final conclusion with a greater level of resilience [27].

Random forest: The Random Forest approach is a type of ensemble learning that merges different decision trees to generate predictions. It follows a two-step process: building the individual trees and making predictions based on the ensemble. In Step 1, individual trees are built by sampling subsets of the training data with replacement, a process known as bootstrap aggregation or 'bagging.' In addition, a random subset of characteristics is considered at each split point to introduce randomness and reduce the correlation between trees [23]. Each tree is grown recursively, selecting the best split at each node based on a parameter like the Gini impurity or the entropy. The process continues until a stopping condition is met. In the second step, the predictions are made independently by each tree. In classification tasks, the final prediction is the class label that gets the most support from the trees. To generate a predicted probability vector for the sample, the total number of votes within each class was divided by the total number of models in the ensemble Random Forest lessens [28]. The gini index of a node n is calculated as follows:

$$Gini(n) = 1 - \sum_{i=1}^{c} (p_i^2) \tag{8}$$

Here, p_i is the proportion of samples in a particular class, and the summation is taken over all classes c_i.

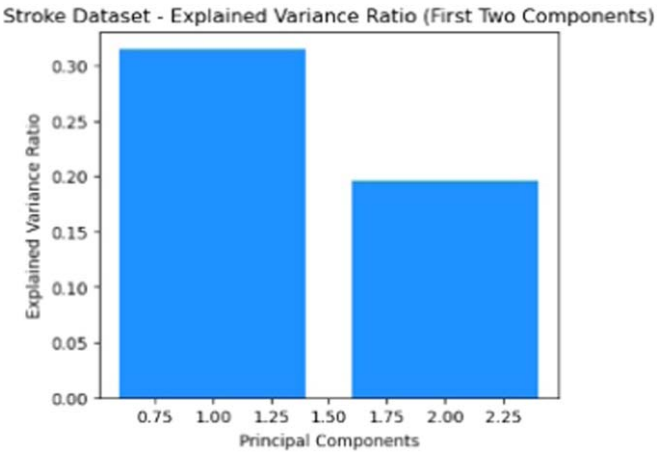


Figure 2. Explained variance ratio of first two components.

Table 2. Analysis of the distinct characteristics' impact on the first and second components. We provide the loading factors' absolute values.

Classifier	PC1	PC2
Gender	0.0002	0.0688
Age	0.4472	0.1938
Hypertension	0.0432	0.0049
Heart disease	0.0435	0.0137
Ever married	0.1444	0.0824
Work type	0.3203	0.4720
Residence type	0.0103	0.0034
Average glucose level	0.7201	0.5710
Bmi	0.3330	0.0945
Smoking status	0.2079	0.6268

Information gain (i) is calculated as follows:

$$InformationGain(i) = parent - \sum(weightedentropy(child) \tag{9}$$

$$parent = -\sum(p_i{*}log2(p_i)) \tag{10}$$

Here, p_i is the proportion of samples in a particular class in the parent node, and the summation is taken over all the classes.

The weighted entropy (child) is the entropy of each child node, weighted by the proportion of samples in that child node.

3.4.2. K-nearest neighbor

The k-nearest neighbors (k-NN) algorithm is the most widely used classification and regression method. During training, it memorizes the feature vectors and their corresponding labels or values. The nearest neighbor k is used to forecast the label for fresh data points during testing. The value of k is chosen, typically an odd number, to avoid ties [29]. The algorithm calculates distances, often using the Euclidean distance between new points and training data. The k nearest neighbor is determined and, for classification, the majority class label among them is assigned as the prediction. The anticipated output in regression is the average of the output values of the k nearest neighbors.

$$Euclideandistance = \sqrt{((x_2 - x_1)^2) + (y_2 - y_1)^2) + ...+(n_2 - n_1)^2)} \tag{11}$$

where $x_1, y_1, \ldots n_1$ and $x_2, y_2, \ldots n_2$ are the characteristic vectors of two data points.

3.4.3. Boosting

Boosting is a machine learning approach that can transform poor learners into a powerful classifier. An ensemble meta-algorithm is employed to mitigate both bias and variation. Meanwhile, a weak learner is a

classifier that exhibits a performance somewhat higher than random guessing, and strong learners are classifiers that achieve high accuracy and serve as a foundation for improving ensemble techniques [30].

The concept of boosting includes applying the underlying learning algorithm to the modified input data iteratively. Boosting techniques train a weak learner with input data, compute predictions, select misclassified samples, and train the next weak learner with an adjusted training set containing misclassified instances from the previous round. Iterative learning continues until a preset number of base learners is obtained and weighted. Boosting reduces bias over variance. Improves high-bias, low-variance base learners. Machine learning may turn poor learners into powerful classifiers. This ensemble meta-algorithm reduces bias and variation. Weak learners are classifiers that perform somewhat better than random guessing, whereas strong learners are accurate and are the foundation of improving ensemble algorithms [31].

eXtreme Gradient Boosting: eXtreme Gradient Boosting (XGBoost)is an ensemble learning algorithm widely used for supervised machine learning tasks. It follows a step-by-step process to create a strong predictive model. Initially, it starts with a simple model that predicts the average of the output labels. Then, it calculates the residuals by comparing the initial predictions with the actual labels. Through gradient boosting, XGBoost iteratively adds new weak-learner models, such as decision trees, to correct the mistakes made by the previous models. Each new model is trained to predict the loss function negative gradient based on the residuals. XGBoost employs an objective function that combines a loss function and a regularization term to measure performance and control model complexity [32]. Build decision trees by recursively dividing the data using features that maximize reduction in the objective function. Pruning techniques are applied to reduce tree complexity. The weak learners are sequentially added to the ensemble and their predictions are combined using weighted averaging. XGBoost incorporates various regularization techniques, including shrinkage, column subsampling, and row subsampling, to prevent overfitting. Once trained, the ensemble can make predictions on new data [33].

3.4.4. Logistic regression

Logistic Regression is used to determine the relationship between one or more independent variables (predictor) and a binary dependent variable (outcome), logistic regression is utilized. As a cost function, the sigmoid function is used. An estimated real value is assigned to a probabilistic value between '0' and '1' by the sigmoid function [34].

$$P(x) = \frac{1}{1 + e^{-x}} \tag{12}$$

Where, P(x) is the probability estimation function with a value between 0 and 1, x is input to the probability function

3.4.5. Stacking

Ensemble learning uses numerous base models to improve prediction performance. The basis model may utilize any machine learning method. An ensemble learning model is homogeneous if it uses a homogeneous set of base learners, otherwise, it is heterogeneous. Ensemble learning uses bagging, stacking, and stacking algorithms. Bagging includes separately learning weak learners and averaging machine learning model predictions. Boosting adds base learners one after another and outputs the weighted average of base model predictions. Other ensemble methods include stacking, which trains basic classifiers on the same dataset and adds meta-learner to enhance model performance [35]. Current research uses the stacking method. The study utilized an architecture that employed two machine learning algorithms as base learners such as XGboost and Random Forest, and a third machine learning algorithm as the meta-learner such as Logistic Regression, Decision Tree and KNN used each of them. The meta-learner utilized models created by two base learners to build the stack-ensemble model necessary for classifying breast cancer recurrence. The meta-learner was utilized to determine the optimal method of combining the predictions from the base learners to get the highest classification accuracy. This work outlines the machine learning algorithms employed and illustrates how a metalearner algorithm will interact with the two input-base learners. The framework used to create the stack-ensemble model in this study is illustrated in figure 3.

4. Results and discussions

In total, there are 5,110 medical records in our collection. Among these data, only 249 relate to individuals diagnosed with stroke, while the remaining 4861 records do not indicate any stroke condition.The data set contains null values. To replace the null values, both mean imputation and interpolation spline curves were used separately. We also compared the classification accuracy of different imputation techniques.

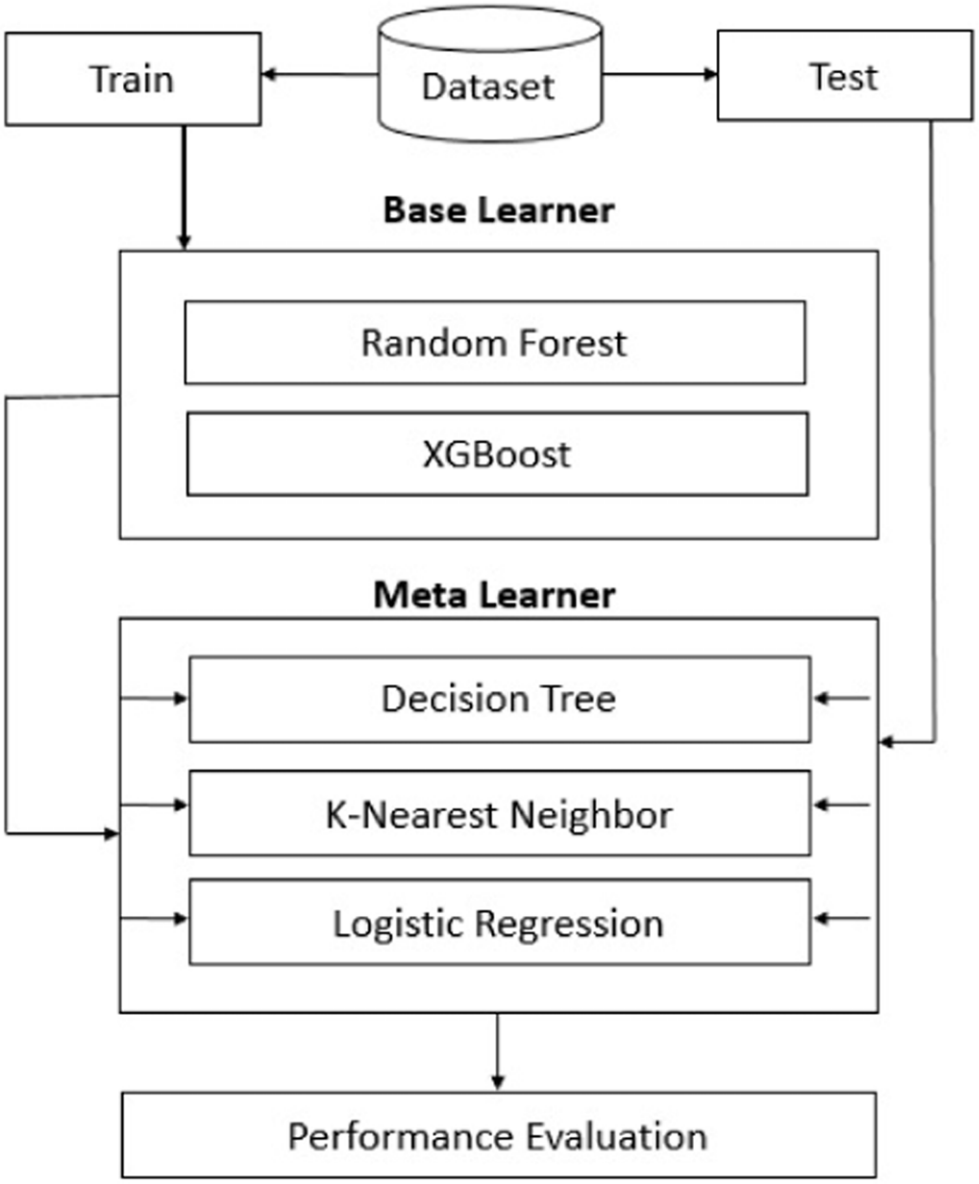


Figure 3. Proposed framework of stacked ensemble.

This dataset has a significant imbalance. This is a challenge when trying to use these data directly for training machine learning models. Consequently, we employ SMOTE (synthetic minority oversampling method) to rectify the imbalance in class distribution by randomly duplicating instances from the minority class. The 249 records are referred to as the minority class, while the remaining 4,861 records without a stroke condition are referred to as the majority class. Afterwards, we compile a dataset that includes 9,722 observations, with 4,861 samples representing the minority and 4,861 samples representing the majority.

We employ Principal Component Analysis (PCA) to analyze the variability in the dataset. The objective of this multivariate analysis is to extract as much variation as possible from the variables by transforming the data set into a set of values of the principal components, a class of linearly independent variables. The primary elements function as succinct representations of the features in the dataset.

In this work, 80% and 20% of the data are used for training and testing, respectively. PCA was used for feature extraction. After applying PCA, seven components were chosen that are fed into the machine learning model and the models' performance was compared based on accuracy score, confusion matrices, recall, precision, F1 score, ROC curve and AUC values.

Since all the models were trained using the same preprocessing methods and PCA of the same number of components, we ensured that each model was trained on an identical dataset. This helped us to keep our experimental setup intact and allowed us to make a proper comparison of model performance.

Table 3. Different performance metrics and accuracy of various base model by using distinct imputation methods without PCA.

Imputation method	Single classifier	AUC value	Accuracy
Mean imputation	Logistic regression	0.86	77%
	Decision tree	0.92	92%
	KNN	0.95	88%
	Random forest	0.98	93%
	XGboost	0.85	93%
Interpolation spline curve	Loistic regression	0.85	77%
	Decision tree	0.89	89%
	KNN	0.94	86%
	Random forest	0.99	94%
	XGboost	0.93	85%

Table 4. Different performance metrics of various base model by using distinct imputation methods with PCA.

Imputation method	Single classifier	Precision	Recall	F1 score
Mean imputation	Logistic regression	0.73	0.79	92%
	Decision tree	0.85	0.91	88%
	KNN	0.91	0.95	93%
	Random forest	0.90	0.94	92%
	XGboost	0.88	0.93	90%
Interpolation spline curve	Loistic regression	0.76	0.84	80%
	Decision tree	0.93	0.98	96%
	KNN	0.93	0.98	96%
	Random forest	0.92	0.98	95%
	XGboost	0.93	0.99	96%

Several performance assessment criteria are used to evaluate the performance of the proposed model. The result is analysed using Confusion Matrix and the accuracy is calculated using the following rule:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

where

TP= Actual results are true and obtained results are also true.

TN=The actual result is true and the obtained result is false.

FP=the actual result is false and the obtained result is also false.

FN=The actual result is false and the obtained result is true.

Recall: The sum of all properly detected positive values divided by the total number of true positive and false negative values is saved. 'True Positive Rate' measures accurately detected positive influences. High recall indicates proper diagnosis.

$$Recall = \frac{TP}{TP + FN} \tag{14}$$

Precision: Precision is determined by the ratio of accurately detected true positive values to the sum of true positive and false positive values.

$$Precisionl = \frac{TP}{TP + FP} \tag{15}$$

F-Measure: The F-measure is derived from recall and precision as follows:

$$F - Measure = \frac{2 * Recall * Precision}{Recall + Precision} \tag{16}$$

Table 5. Different performance metrics and accuracy of various meta classifiers by using distinct imputation methods using PCA.

Imputation method	Meta learner	Precision	Recall	AUC value	Accuracy
Mean imputation	Decision tree	0.92	0.97	0.95	95%
	KNN	0.90	0.97	0.95	95%
	Logistic Regression	0.90	0.97	0.99	93%
Interpolation spline curve	Decision tree	0.93	0.99	0.95	95%
	KNN	0.95	0.98	0.95	96%
	Logistic Regression	0.96	0.98	0.99	97%

Table 6. Comparison of the specificity, sensitivity, and accuracy of single classifier and stack ensemble classifier by using interpolation spline curve imputation with PCA features.

Classifier	Model	Specificity	Sensitivity	Accuracy
Single Classifier	Logistic Regression	72%	84%	78%
	Decision tree	93%	98%	95%
	Random forest	91%	98%	95%
	KNN	93%	98%	95%
	XGboost	93%	99%	96%
Stacking with various meta learner	Decision tree	92%	99%	95%
	KNN	95%	98%	96%
	Logistic regression	96%	98%	97%

Table 7. Comparison table of proposed work with other work.

References	Database	Filling Missing value	Method	Accuracy in (%)
[12].	Kaggle	Mean	Voting classifier	91
[36]	HealthData.gov	Random forest regression	DNN	71.6
[37]	kaggle	NA	PNN	95
[38]	Kaggle	Mean	NB,DT,RF	89.9, 91.9 and 94.9
[39]	kaggle	KNN	RF,XGBoost, LightGBM	96.34
Proposed Method	**Kaggle**	**Interpolation Spline curve**	**Stack Ensemble Learner**	**97**

Table 3 shows the different metrics results of various baseline techniques and different imputation methods without PCA. We have observed that by using mean imputation, Random Forest and XGboost both gives 93% accuracy. On the other hand, Random Forest performance slightly improved by employing an interpolation spline curve. And achieved 94% accuracy without PCA.

Table 4 presents the performance metrics of various base models applying different imputation strategies combined with PCA.

Table 5 shows the various performance metrics with stack ensemble approach using PCA. We have observed that the accuracy of Random Forest and XGBOOST as a base classifier and Decision Tree, KNN, and Logistic Regression as meta-learner are 95%, 96% and 97%, respectively. The stacking ensemble approach uses the bagging and boosting capabilities by combining them, resulting in more reliable and precise predictions compared to using single models alone. Implementing this greatly improved the accuracy of predicting the occurrence of brain stroke. Upon comparing the precision, recall, F1 score and AUC scores, the highest precision achieved by the logistic regression as a meta-learner is 0.96. The AUC scores for KNN, decision tree, and Logistic Regression classifier are 0.95, 0.95, and 0.99 respectively.

Table 6 demonstrates the effectiveness of the Logistic Regression meta-learner in achieving high accuracy (97%), precision (0.96), and recall (0.98). While its specificity is slightly lower than other classifiers (e.g., Decision Tree and KNN), this trade-off is expected in models that prioritize sensitivity to avoid false negatives in

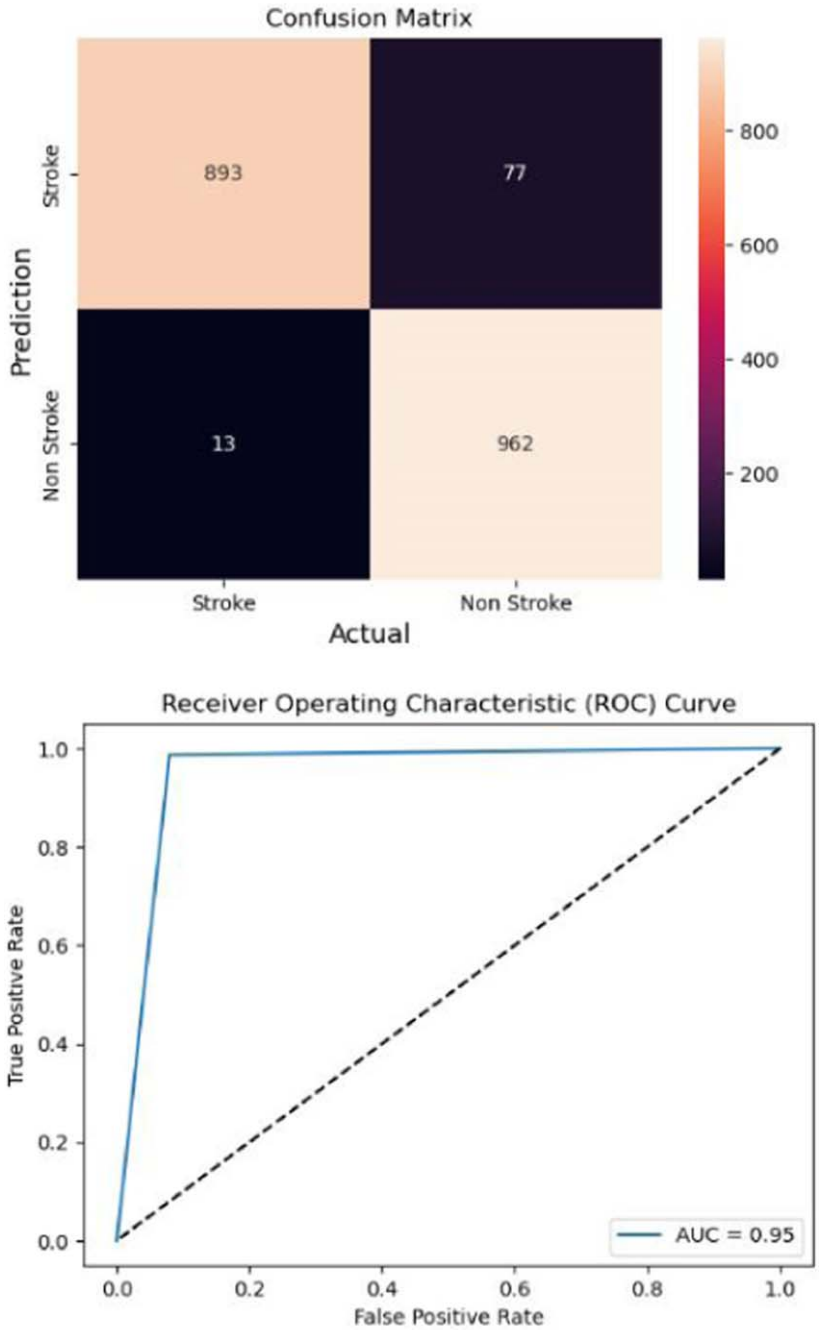


Figure 4. Confusion matrix and ROC curve of ensemble classifier where decision tree used as meta learner.

critical applications like stroke detection. Table 7 shows the comparison of proposed work with others existing works.

Figures 4, 5 and 6 illustrate the confusion matrix and the ROC curve for several different ensemble classifiers.

This study uniquely outlines several endeavors that appear exceptional and potentially influential in the current body of studies. Specifically, there are two salient points that make the paper very inventive. The use of interpolation spline curves in data imputation renders the technique most adaptable and high-resolving throughout the process. The technique's performance is markedly superior when employing a holistic approach, compared to the often-utilized isolated technique that focuses on individual variables, such as mean imputation of missing data using k-Nearest neighbors or other convenient techniques. All this was achieved through Principal Component Analysis prior to the model building procedure, which allows the elimination of redundant information while preserving principal data that could increase the predictive power of the model. The experiment uses a stacked ensemble technique that combines a number of classifiers, which were employed as follows: Random Forest, XGBoost, and Logistic Regression. Its superior performance in prediction capability

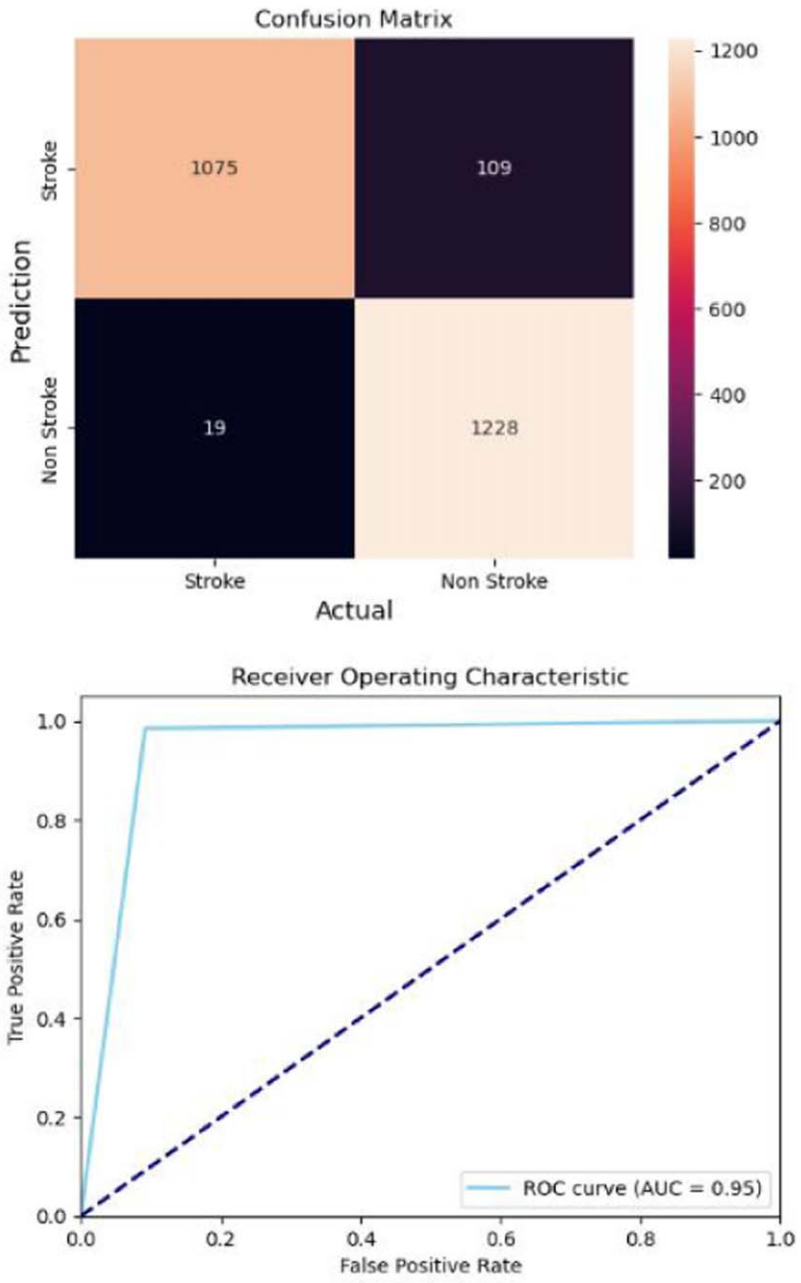


Figure 5. Confusion matrix and ROC curve of ensemble classifier where KNN used as meta learner.

with an accuracy of 97%, precision of 0.96, and recall of 0.98 is achieved. The practice also effectively uses SMOTE to resolve class imbalance problems, which are frequently encountered with medical data. Importantly, the model can effectively predict stroke occurrence. It also comes with the comparison of the present method against some benchmark models, demonstrating the incremental advantages of the use of the present technique in terms of both the prediction model developed and its robustness. However, there are constraints which have been happening and which the research is comfortable dealing with. For example, the data used are from Kaggle, which may not have the actual complexity of health data and has no source code available, thus affecting the transparency as to how the outcomes have been arrived at. The consideration of ethics and precautionary control plans such as protection of data and discrimination are touched on, but very strongly since they infertility. Nevertheless, this investigation is a good point for long-stroke prediction and also advances modular design approach for wide range of real-world development. Considering that the paper does not address issues of ethics and practical implementation measures, including data protection and fairness, in practice, only a few observations are necessary, since more comprehensive information is required. However, this work, even with its long-term strategy, prepares the technology for research on stroke prediction, with functionalities that can be embedded in real-world situations.

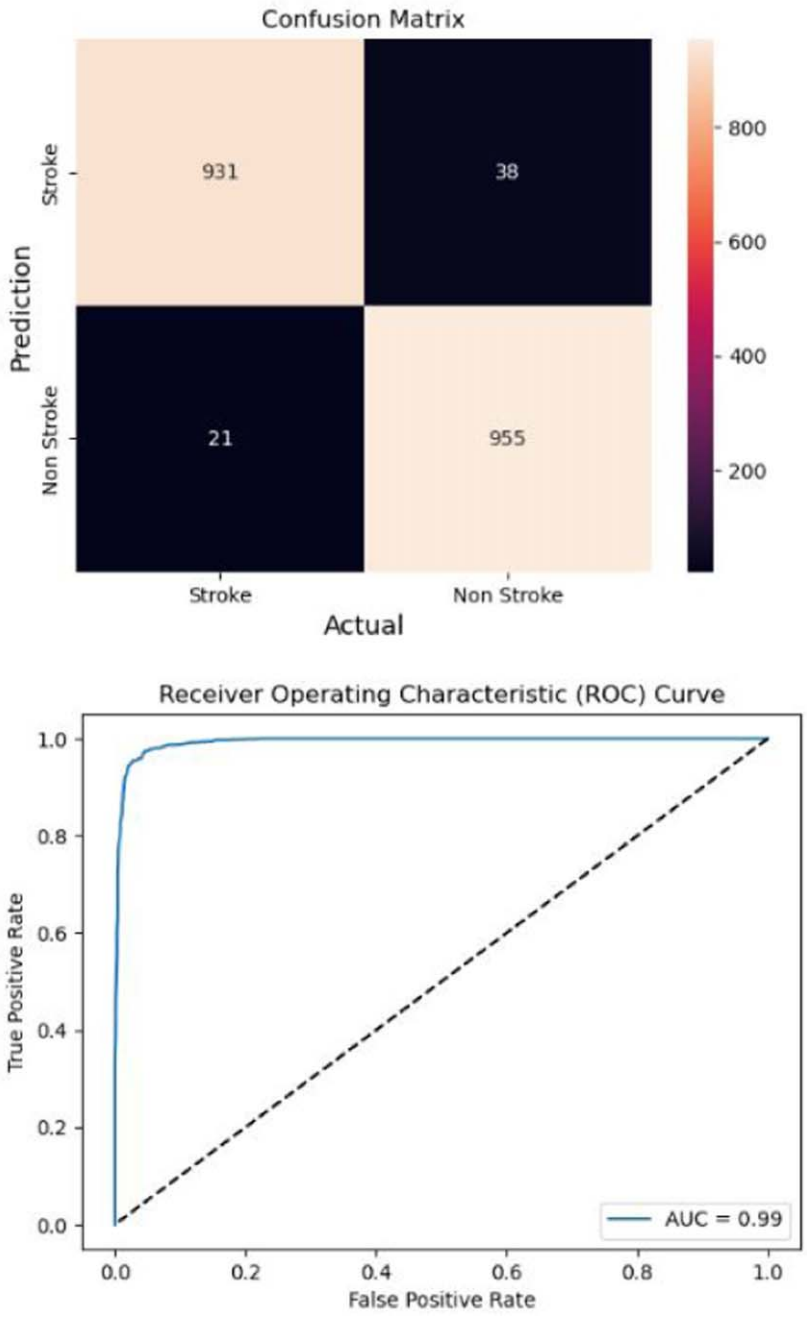


Figure 6. Confusion matrix and ROC curve of ensemble classifier where Logistic Regression used as meta learner.

5. Conclusion

This paper discussed the necessity of stroke identification. Our strategy for achieving this goal is to create a system that handles the missing data problem during the preprocessing stage by employing the interpolation spline curve model, which we demonstrate is the ideal method for filling missing values. In addition, we conducted a principal component analysis to identify the most important factor in the identification of stroke. The research revealed that nearly all of the main components are necessary to account for greater variation. However, variable loadings revealed that the largest variance component, the first, may explain the underlying phenomenon of stroke prediction. Robust Stacking Ensemble Learner (SEL) is developed using ensemble learning to maximize precision. The proposed model tested electronic health records for brain stroke taken from Kaggle. Here, we compare the performance of various baseline techniques with the stack ensemble classifier. According to the results obtained, it is concluded that the Decision Tree and K-Nearest Neighbor as metalarners show their robustness in ensemble machine learning model. The observation from the result shows that the interpolation spline curve performs well for data imputation, and the ensemble classifier used Random Forest

and XGBoost as a base learner with Logistic Regression as a meta-learner by using PCA features providing better performance.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://kaggle.com/datasets/fedesoriano/stroke-prediction-dataset [23].

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